POAMS: LECTURE 17
MORE ON POAMS AND ORBITAL MOTION

In this shorter lecture, we shall continue to study constrained and natural circular orbits in POAMS and I will provide a further numerical example. I will then move on to discuss the POAMS analogue of Newton's Second Law for natural orbital motion. Again, this lecture is based on some of Chapter 6 of 'Light-speed….'.

1. The two-body problem and the POAMS equation of motion [*Recap*]

In this lecture, we continue to apply POAMS to the study of the two-body problem. Recall that this consists of a particle, *P*, of mass *m* orbiting a body, *B*, of mass *M*, with *m* very much less than *M*. As before, we let the centre-of-mass of *B* lie at the origin, *O*, of a set of Cartesian coordinates and let the position vector of *P*, in relation to *O*, at any particular time *t* be $\mathbf{r}(t)$ [*see diagram given in the previous two lectures*].

Recall that, based on conservation of momentum and the empirical evidence alone, the POAMS second-order equation of motion for *P*'s orbit about *B* for which stable closed orbits are possible is,

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - \frac{L^2}{m^2 r^3} = -\frac{\alpha}{r^2} \qquad (1)$$

where $r(t) = \|\mathbf{r}(t)\|$, L is the constant magnitude of the angular momentum of *P* about *B,* and α is a constant [*Lecture 15, equation (11)*]. This includes the special cases of 'radial free-fall' when $L = 0$, and circular motion, for which *r* is a constant.

Since we wish to compare equation (1) with the corresponding Newtonian equation of motion [*Lecture 13, equation (9)*], in any *natural orbit* of *P* about *B*, we re-label the constant α as *GM*, where *M* is the mass of *B* and *G* is simply a new constant, with no 'gravitational' connotations.

Recall that, according to the POAMS thesis, the true (*i.e.* measurable) force on the particle *P* is not *concealed* in its natural orbit but is *revealed* in the difference between the natural orbit and an 'unnatural' one, that is, some state of motion in which the particle is held by a measurable force, of

magnitude F. In the specific case of a constrained orbit of P with an equation of motion of the form (1), we may let $\alpha = G'M$, where G' is a constant with the same dimensions as the Newtonian 'gravitational' constant, G, but only equal to G in any natural (unconstrained) orbit of P. Then the measurable force exerted on P in any constrained orbit has magnitude, F, given by

$$F = |G - G'|\frac{mM}{r^2} \qquad (2)$$

[*Lecture 16, equation (6)*]. Otherwise, in a natural, 'force-free' orbit, $G' = G$ and $F = 0$.

---

2. Circular orbits [*Recap*]

For now, we continue to restrict our discussion to the special case of *circular* orbits, both natural and constrained. Such orbits occur when r is a constant in equation (1) and a particular value of r gives a particular circle of radius r. In this special case

$$L = mvr \qquad (3)$$

[*Lecture 16, equation (7)*], where v is the constant orbital speed of P. It then follows by equation (1) with $\alpha = GM$, that any natural circular orbit satisfies

$$v^2 = \frac{GM}{r} \qquad (4)$$

[*Lecture 16, equation (8)*], so that there is a *unique* natural circular orbit for any given radius. Any *constrained* circular orbit about Earth satisfies equation (4) with G replaced by a new constant G'. [*Lecture 16, equation (9)*].

Given a constrained orbit of P with constant angular momentum of magnitude L, there is a unique *circular natural* orbit for P, with the same magnitude of angular momentum, where from equations (3) and (4), the orbital speed, v_0, of P is given by

$$v_0 = \frac{GmM}{L} \qquad (5)$$

and the radius, r_0, of P's orbit is then given by

$$r_0 = \frac{L}{mv_0} = \frac{L^2}{m^2GM} \qquad (6)$$

[*End of recap*]

In the example that follows, the figures given are intended only to illustrate the key ideas. The constants quoted are the currently accepted values and differ slightly from the figures used in 'Light- Speed…'.

3. Example Three

In Example Two of the previous lecture, we studied an object that was fixed on the surface of the rotating Earth. We now consider a rocket-propelled road vehicle, P, of mass m = 1400 kg say, which is capable of travelling at a substantial constant speed. Note that the actual choice of the mass of P does not affect the following argument. If initially, P is at rest somewhere near Earth's equator, then it follows from Example Two that P will press on Earth's surface with a force of magnitude
$F \cong 9.80665 \times 1400 \text{ N} = 13729 \text{ N}$, and hence weighs 1400 kg.

Now suppose that the vehicle P is set in motion and travels westwards along a track at constant speed V = 300 m s^{-1} say (*i.e.* approximately 675 mph). Then P counteracts Earth's rotational speed to some extent and, in effect, its constrained orbital speed relative to the centre of Earth is
$v \cong 465.111 - 300 \text{ ms}^{-1} = 165.111 \text{ ms}^{-1}$. (Earth rotates in a *prograde* direction, *i.e.* from west to east and so rotates anticlockwise relative to the North Pole.) With this value of v, it follows by equation (4) with G replaced by G', and with $r \cong 6.378137 \times 10^6$ m and
$M \cong 5.997939 \times 10^{24}$ kg [*see Example Two, Lecture 16*] that

$$G' \cong 2.898975 \times 10^{-14} \text{ m}^3 \text{ kg}^{-1}\text{s}^{-2}$$

This means that according to equation (2), with $G \cong 6.6743015 \times 10^{-11}\text{m}^3\ \text{kg}^{-1}$, that *P* would press on the track with a force of magnitude $F \cong 13771$ N. Hence, the vehicle weighs *more* when it is moving westwards then when it is stationary relative to the surface of Earth. This is borne out by the fact that, using equation (3), the moving vehicle has an effective angular momentum of

$$L \cong m \times 1.053101 \times 10^9\ \text{kg m}^2\ \text{s}^{-1}$$

Then, by equation (6), the vehicle would seek to follow a natural circular orbit, if allowed, of radius $r_0 \cong 0.277033 \times 10^4$ m, which is less than its natural orbital radius given in Example Two, for the case when it is stationary on Earth's surface [note that $r_0 \cong 2.198331 \times 10^4$ m in Example Two.]

Now suppose that *P* travels in an easterly direction along the track, and hence in the same direction as that of Earth's rotation, again at a speed of $V = 300\ \text{m s}^{-1}$. The effective speed of *P* relative to Earth's centre is now $v \cong 465.111 + 300\ \text{ms}^{-1} = 765.111\ \text{ms}^{-1}$. In this scenario, it follows by equation (4) with *G* replaced by *G′*, that

$$G' \cong 6.225021 \times 10^{-13}\ \text{m}^3\ \text{kg}^{-1}\text{s}^{-2}$$

This means, according to equation (2), that *P* would press on the track with a force of smaller magnitude $F \cong 13648$ N. Hence, the vehicle weighs *less* when it is moving eastwards then when it is stationary relative to the surface of Earth.

This decrease in the weight of the vehicle *P* when moving in the easterly direction, in combination with the rotation of Earth, fits the description of *levitation* of *P* with respect to Earth. This is the result of combining the angular momentum of the vehicle due to Earth's rotation with the angular momentum of the vehicle itself due to energy supplied by its own engine. As a consequence of this additional angular momentum, the measurable force between the two bodies is decreased. Again, this is supported by the fact that, using equation (3), the moving vehicle now has an effective angular momentum of

$$L \cong m \times 4.879983 \times 10^9\ \text{kg m}^2\ \text{s}^{-1}$$

Then, by equation (6), this time the vehicle would seek to follow a natural circular orbit, if allowed, of radius $r_0 \cong 5.948793 \times 10^4$ m, which is more than its natural orbital radius given in Example Two, in the case when it is stationary on Earth's surface.

---

4. <u>The POAMS analogue of Newton's Second Law</u>

In the case of a particle *P* orbiting a body *B* of mass *M*, it follows by the Newtonian/POAMS equation of motion (1), with $\alpha = GM$, that since there is no implied *in vacuo* 'gravitational' force determining a *natural* orbit of *P*, there is also no implied centrifugal force, $m\mathbf{a}$, felt by the particle *P* in such an orbit. In other words, although *P* has an acceleration **a**, which is non-zero at each point of its orbit, any *natural* orbit of *P* is 'force free', so that the vector quantity $m\mathbf{a}$ is not a real, measurable force in the classical sense. This implies that in POAMS, we require an amended version of Newton's Second Law.

In POAMS, then, we require a new definition of an acceleration, which is zero in any natural, force-free, orbit. Recall, from equations (2) and (3) of Lecture 16, it follows that the orbital acceleration, $\mathbf{a}(t)$, of *P* is given by

$$\mathbf{a}(t) = \left(\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - \frac{L^2}{m^2 r^3}\right)\mathbf{n}(t) \quad (7)$$

where $\mathbf{n}(t)$ is a vector of unit length in the direction of *P*'s position vector, $\mathbf{r}(t)$. Hence, from equation (1), with $\alpha = GM$, it follows for any *natural* orbit of *P* that,

$$\mathbf{a}(t) = -\left(\frac{GM}{r^2}\right)\mathbf{n}(t) \quad (8)$$

The obvious course, then, is to define the **natural acceleration**, $\mathbf{a}^*(t)$, for any orbit of the particle *P* in our two-body system as

$$\mathbf{a}^*(t) = \mathbf{a}(t) + \left(\frac{GM}{r^2}\right)\mathbf{n}(t) \quad (9)$$

where, as above, $\mathbf{a}(t)$ is the classical acceleration of P and $\mathbf{n}(t)$ is a vector of unit length in the direction of P's position vector $\mathbf{r}(t)$. It then follows by equations (8) and (9) that $\mathbf{a}^* = \mathbf{0}$ along any natural orbit of P, whereas, along any other orbit of P, using equation (8),

$$\mathbf{a}^*(t) = (G - G')\frac{M}{r^2}\,\mathbf{n}(t) \quad (10)$$

Notice that when $M = 0$, $\mathbf{a}^* = \mathbf{a} = \mathbf{0}$. In this case, it follows by equation (1) that any natural orbit of P is a straight line, which means that in POAMS, rectilinear motion occurs as a special case of orbital motion. This makes Newton's First Law as a special case of the conservation of angular momentum, which also implies conservation of linear momentum.

Having defined the natural acceleration of an orbiting particle, we may, then, define the **natural** or **intrinsic force**, $\mathbf{F}^*$, felt by the particle P of mass m in any orbit about the body B of mass M by

$$\mathbf{F}^* = m\mathbf{a}^* \quad (11)$$

where $\mathbf{a}^*$ is the natural acceleration of P. Equation (11) is the POAMS analogue of Newton's Second Law. From this it follows that $\mathbf{F}^* = \mathbf{0}$ in any natural orbit of P and that $\mathbf{F}^* \neq \mathbf{0}$ otherwise. It follows from equations (10) and (11) that along any constrained orbit of P for which angular momentum is conserved,

$$\|\mathbf{F}^*\| = |G - G'|\frac{mM}{r^2}$$

so that it follows by equation (2) that the magnitude of $\mathbf{F}^*$ is the real, measurable 'gravitational' force on P in any constrained orbit.

In these first lectures on POAMS, I have claimed that, from the standpoint of Normal realism, the classically conceived so-called 'gravitational constant' can be treated as a variable. I have shown that, in POAMS, the orbits of freely-moving particles are explained in terms of angular momentum alone, together with empirical evidence, with no need for the concept of an 'in vacuo force'. Beginning in the next lecture, I

discuss, again in Normal Realist terms, the effect of spin on the orbital parameters of free-moving particles and hence, on the value of G.

Anthony D. Osborne, March 2024.